Types of Numbers

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Types of Number

You will come across vocabulary such as

- **Integers** and **natural** numbers
- **Rational** and **irrational** numbers
- **Multiples**
- **Factors**
- **Prime** numbers
- **Squares**, **cubes** and **roots**
- **Reciprocals**

Knowing what each of these terms mean is essential.

What are integers and natural numbers?

- **Integers** are **whole** numbers;
  - They can be **positive, negative** and **zero**
  - For example, -3, -2, -1, 0, 1, 2, 3 are all integers
- **Natural** numbers are the **positive integers**
  - They can be thought of as counting numbers
  - 0, 1, 2, 3, 4, ... are the natural numbers
    - Notice that 0 **is** included

What are rational and irrational numbers?

- A **rational** number is a number that can be written as a fraction in its simplest form
  - A rational number can be written in the form $\frac{a}{b}$
    - where a and b are both **integers,** and b is not zero
  - All **terminating** decimals are rational numbers
    - e.g. $0.32 = \frac{32}{100}$
  - All **recurring** decimals are rational numbers
    - e.g. $0.4444... = \frac{4}{9}$
    - Recurring digits are indicated by dots or bars
      e.g. $0.4\dot{5} = 0.4\overline{5} = 0.455\ 555\ 555\ ...$
      e.g. $0.\dot{5}82\dot{4} = 0.\overline{5824} = 0.5824\ 5824\ 5824\ ...$
- An **irrational** number cannot be written as a fraction

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- It has an endless amount of decimal places and there is **no pattern** or recurring digits
- Any (simplified) fraction involving π, such as $\frac{\pi}{2}$, will also be irrational (as a in $\frac{a}{b}$ is not an integer)

Examiner Tips and Tricks

- In the calculator paper, use your calculator to your advantage!
  - To determine whether a number is rational or irrational, type it into your calculator and see if it can be displayed as a fraction (with integers)

Worked Example

Explain why $0.\dot{3}49\dot{2}$ is a rational number.

The dots indicate the first and last of the recurring digits

$$0.\dot{3}49\dot{2} = 0.3492\ 3492\ 3492\ \ldots$$

Answer:

$0.\dot{3}49\dot{2}$ is a rational number as it is a recurring decimal

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